Circuit for controlling the concentration of CO

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Input data

Input voltage: 11V

Sensor Resistance: 35-5kΩ

LED color: Orange

The measurement range of the sensor: 400-9000 [ppm]

Indoor CO concentration: 400-6000[ppm]

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1 Introduction

1.1 Requirements

The objective of the project is to design a system that uses a resistive gas sensor to maintain the concentration of carbon monoxide within a specified range in an enclosure. When the concentration has reached the upper limit of 6000 ppm, the system will activate a fan that introduces fresh air. Conversely, when the concentration drops to the lower limit of 400 ppm, the system turn off the fan, so that the concentration of carbon monoxide always remains within the desired range. The resistance of the sensor varies from 35kΩ at 400 ppm to 5 kΩ at 9000 ppm. This variation in electrical resistance of the sensor must be converted into a voltage variation in the range of 2V to 9V. The fan is controlled by a hysteresis comparator through a relay, which is modeled with a resistor.

In what follows, I will present the block diagram which shows my solution for the presented requirement:

1.2 Block Diagram

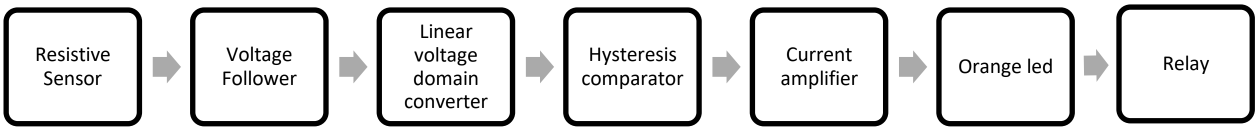

Figure 1- Block Diagram

Figure 1- Block Diagram is a graphical representation of the carbon monoxide detection system, illustrated by functional components and their interconnection. Each block represents a component within the system, the ones that I used in order to complete the given task. It is a sketch of the project, that breaks the functionality of the circuit in smaller "blocks", facilitating the understanding of the flow of the configuration.

1.3 Description of the functionality of each block

The ***resistive sensor*** is formed by a variable resistor defined by {Rs}, a parameter that varies between 5kΩ and 35kΩ and a fixed resistor that has a value close to the higher limit of the interval, in order to have a voltage divider between that resistor and the sensor, both of them being connected to Vcc (11V).

The ***voltage follower*** has the role of stabilizing the signal that exits the voltage divider, in order to provide the next block of circuit a clean signal.

The linear ***voltage domain converter*** has the role of converting the initial voltage provided by the output of the voltage follower to the domain of [2÷(Vcc-2)]V, that was specified in the requirement.

The ***hysteresis comparator*** provides the necessary range of values in which the sensor should detect the quantity of dangerous gas in the enclosure, having the aim of opening the fan on time, providing an acceptable quantity in the enclosure.

The ***current amplifier*** is used for amplifying the current at the output of the hysteresis comparator, because the relay needs a large quantity of electric current in order to operate, meaning that without amplification it would burn the operational amplifier of the comparator.

The ***orange led*** is a tool that assures visibility of the functionality of the circuit, due to the fact that it is turned on whenever the quantity of gas is not in the required interval, signalizing a problem. It is opened as long as the fan is operating. The color is a simple specification of design.

The ***relay*** is modeled with a resistor that uses the real value of the component, such that the simulation is as precise as possible.

1.4 Circuit Schematic

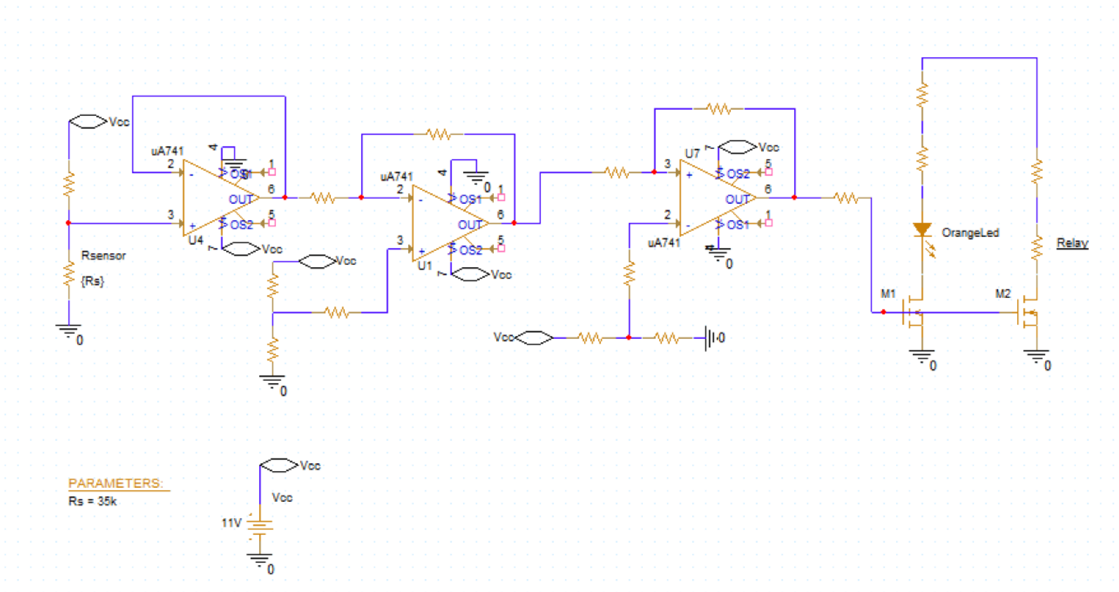

Figure 2 - Electrical Schematic

The electrical schematic is a representation of the block diagram, in which I combined each block to build a circuit. At this point, its components do not have values because I will present in the following chapters the dimensioning procedure for each functional block of the circuit.

1.5 Theoretical aspects regarding the components used

1.5.1 Operational amplifier

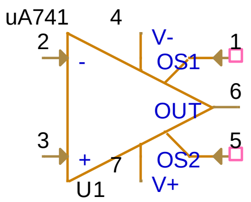

Figure 3 – Operational Amplifier

The Operational Amplifier I chose to work with is uA741. This is a general-purpose operational amplifier that is widely used in various analog application. There are multiple reasons why I wanted to use it as one of the main components of my design, one of the most significant one being its offset-voltage null capability, allowing for precise adjustments, but also its protection against short circuits and the absence of the latch-up, fact that assures stable operation. It is widely used for voltage followers and general feedback circuits, which makes it a basic amplifier, not the best solution, but suitable for the present application, having proper values for this project, such as:

- the uA741 can operate with a wide range of supply voltage typically from ±10V to ±18V, which makes it suitable for this application, for which I used 11V ;
- despite of the fact that the slew rate has a very small value, 0.5 V/µs, it is still usable for analog applications, including those requiring DC precision (my particular project being such a circuit) and moderate-speed signal processing;
- the uA741 has internal protection, to prevent damage from short circuits, which enhances reliability and longevity in practical applications.

1.5.2 Resistors

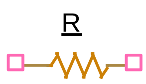

Figure 4 Resistor

A resistor is a passive two-terminal component that opposes the flow of electric current. The values I chose from each resistor will be presented in **Chapter 2**.

1.5.3 DC current source

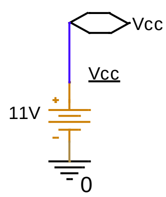

Figure 5- DC current source

The DC voltage source produces a constant voltage output. This element is essential for supplying the circuit, providing 11 V, just as specified in the input data provided in the requirement.

1.5.4 Orange led

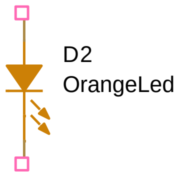

Figure 6 - Orange led

An LED(light-emitting diode) is a semiconductor device that emits light when an electric current flows through it.

The LED I chose has the name LO-Y876-TYP, this one being a model of an orange LED that I found in OrCAD Library. This led has a voltage of 2V and it needs 20mA to turn on.

1.5.5 Transistor

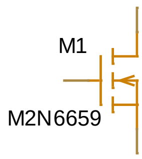

Figure 7 - nMOS transistor

The nMOS transistor is a three-terminal device used for switching and amplification. In an N-channel MOSFET, the majority carriers flow through the channel between the source and drain terminal.

The reason why I specifically choose the M2N6659 transistor is due to the high current handling capability (up to 4.5A), which is crucial when driving inductive loads, like relays, that require more current than typical logic-level that the operational amplifier in Figure 3 provides (20mA), in order not to have a current of a such high value that it burns the hysteresis comparator. Aswell, this transistor is appropriate for the given application, because of the low on-resistance, meaning that it can conduct large currents with minimal voltage drop across the drain-source channel when turned on, reducing the power dissipation and increasing efficiency.

2 Calculus and Simulation Results for initial circuit

In what follows, I will explain all the components added in the initial circuit (the one before standardization) and the results of the simulation, in order to have a term of comparation with the standardized circuit presented in chapter x.

2.1 Schematic of the non-standardized circuit

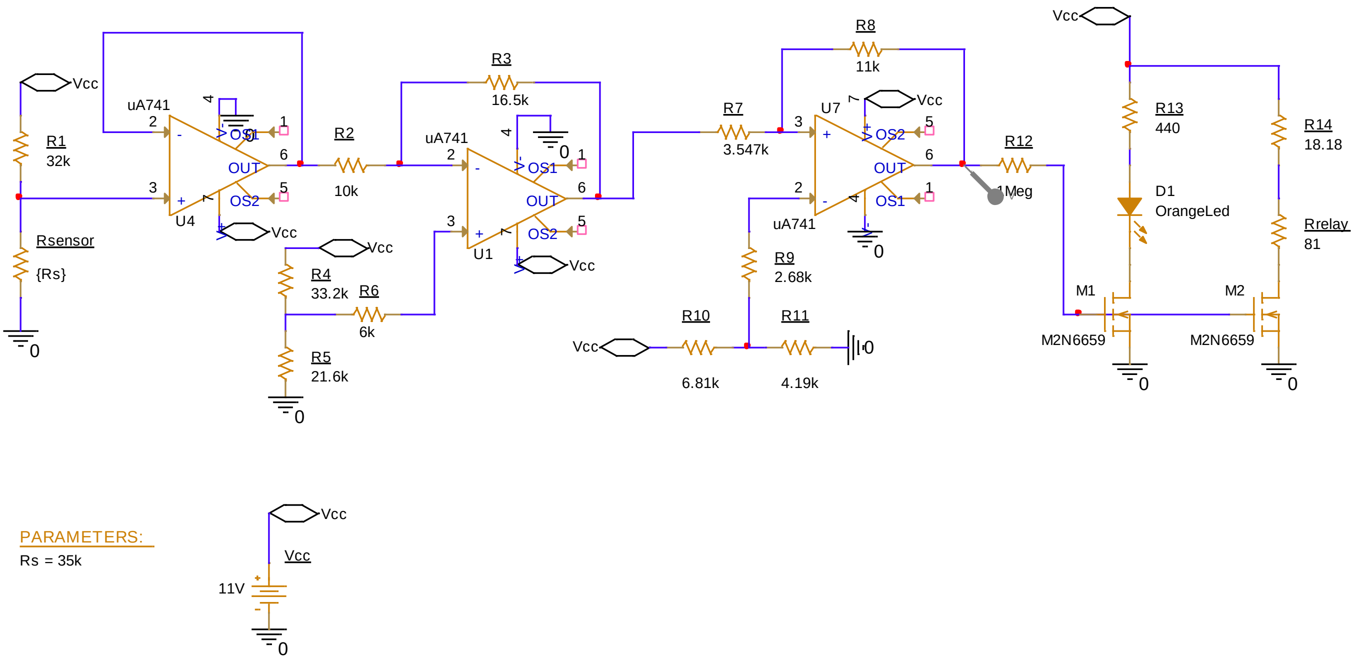

Figure 8 - Schematic of the non-standardized circuit

The schematic corresponds to the block diagram (Figure 1). In what follows, I will present each part of the schematic, the calculus and the simulations.

2.2 Description of the elements of the circuit

2.2.1 Resistive sensor

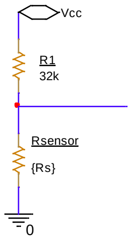

Figure 9 -Voltage divider

The sensor is modeled by a voltage divider, consisting of the variable resistor of the sensor and a high resistance component that assures a voltage range in which the sensor operates as follows:

$$v_{R_{sensor\ min}} = \frac{R_{sensor\ min}}{R_{sensor\ min} + R1} \cdot Vcc = \frac{5}{5+32} \cdot 11 = 1.48V \quad [1]$$

$$v_{R_{sensor\ max}} = \frac{R_{sensor\ max}}{R_{sensor\ max} + R1} \cdot Vcc = \frac{35}{35+32} \cdot 11 = 5.74V \quad [2]$$

This can be proven by a simulation in OrCAD, which shows that, using a DC sweep that varies the sensor resistance between 5kΩ and 35kΩ as in Figure 10, one can observe that the above computed range value (equations [1] and [2]) is exactly as the real one, presented in Figure 11.

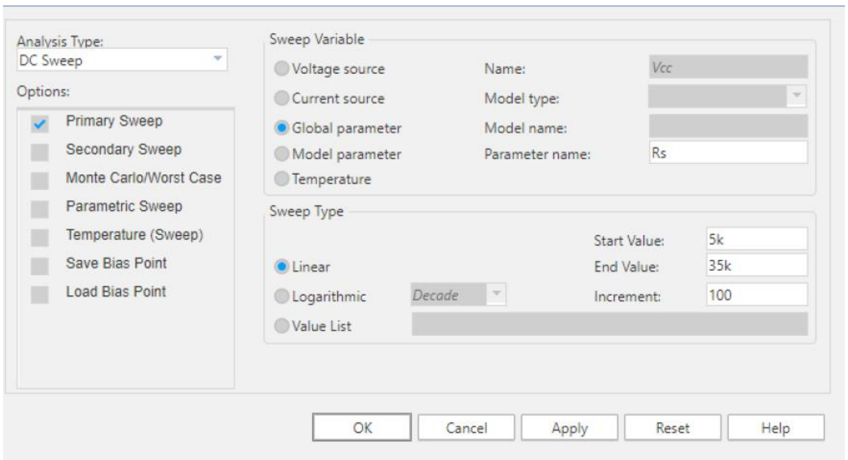

Figure 10 - Simulation profile for voltage divider

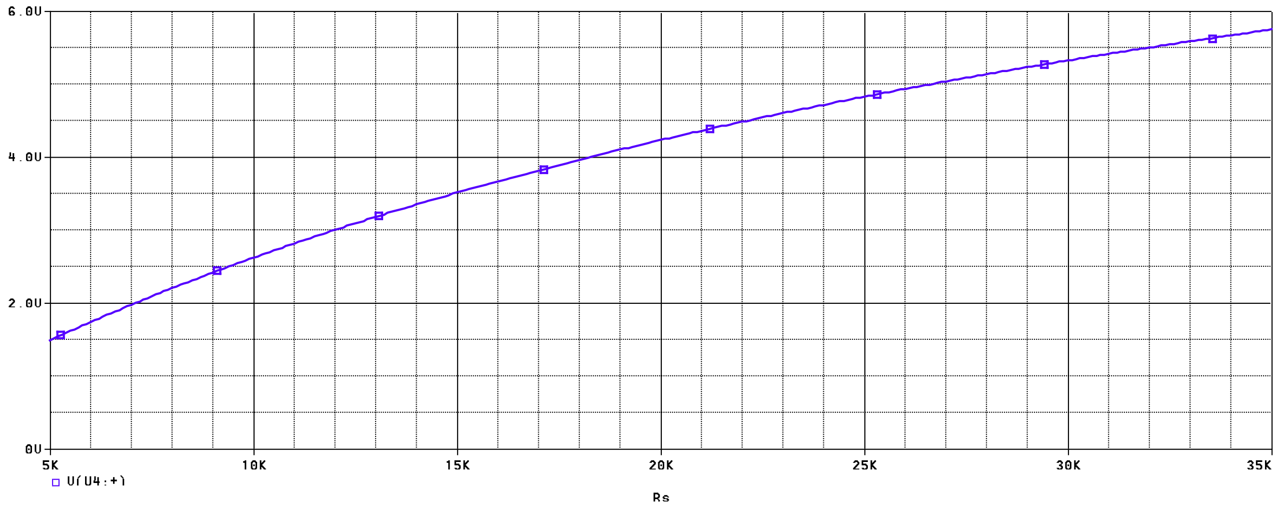

Figure 11 - The voltage characteristic at the output of the resistive sensor

Placing cursors on the lower and higher limit of the graph at Figure 11, it is even more obvious that the theoretical assumption corresponds to the simulation.

Trace Color	Trace Name	Y1	Y2
	X Values	5.0000K	34.900K
CURSOR 1,2	V(U4:+)	1.4862	5.7371

Figure 12 - Values shown by cursors on voltage divider output characteristics

2.2.2 Voltage Follower

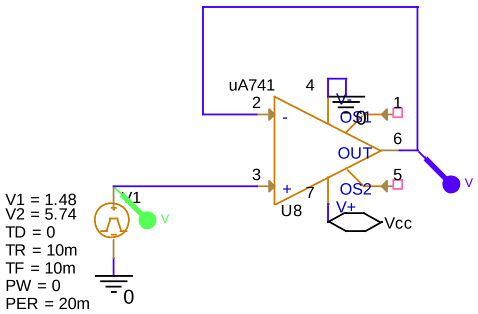

Figure 13 - Voltage follower

The voltage follower, also known as a buffer amplifier, is an op-amp circuit with a voltage gain of unity. The purpose of this circuit is to act as a buffer, isolating the sensor from the subsequent stages of the circuit. This isolation ensures that any changes in the load connected to the output of the voltage follower do not affect sensor's operation, maintaining the integrity of the sensor signal. In addition, it is used for impedance matching between the high impedance sensor(R sensor) and the rest of the circuit. By presenting a high impedance to the sensor, the voltage follower ensures that the sensor's signal is not loaded down, preventing attenuation of the signal.

In what follows, I will present through a time domain analysis, the result of this segment of circuit, to which I added a Vpulse source that provides the voltage follower the voltage at the output of the sensor, so that we can observe that this circuit does not change the input signal provided by the sensor in any way, as seen in Figure 16 .

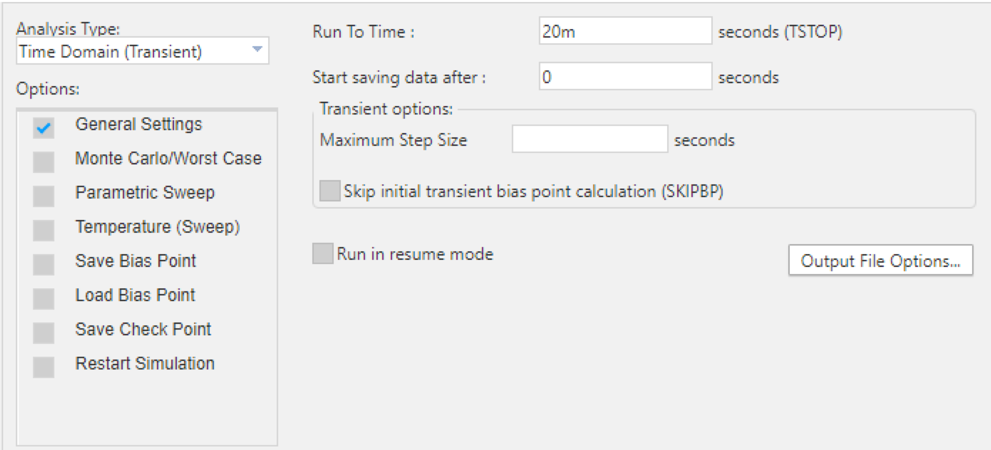

Figure 14 - Simulation profile of time domain analysis

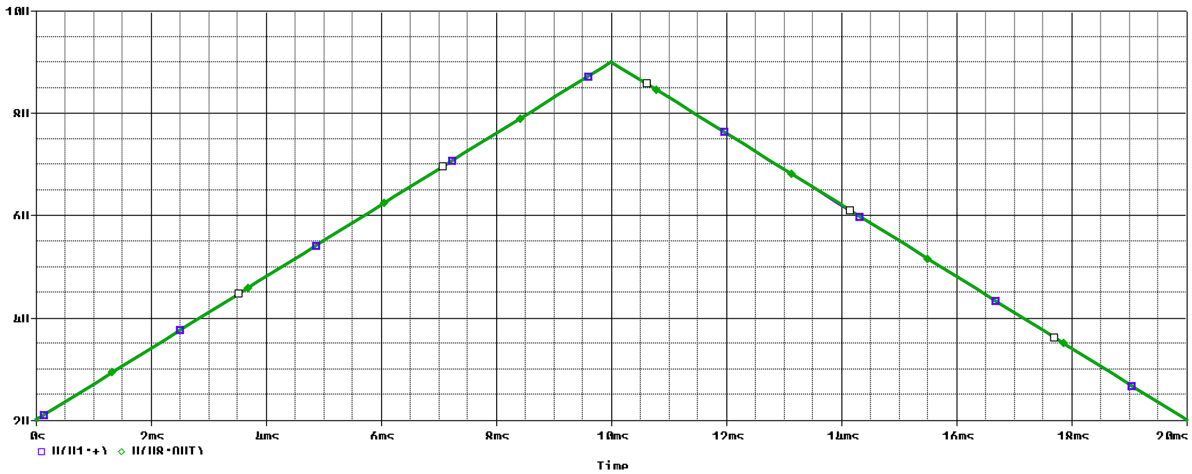

Figure 15 - Output and input voltage of the voltage follower

Trace Color	Trace Name	Y1	Y2
	X Values	0.000	10.000m
CURSOR 1,2	V(U8:-)	1.4802	5.7399
	V(V1:+)	1.4817	5.7400

Figure 16 - Values of the input and output voltage of the voltage follower

2.2.3 Linear Voltage Domain Converter

An inverting voltage domain conversion circuit, presented in Figure 17, is a circuit with an op-amp that can convert a command voltage into a new range of values at the output.

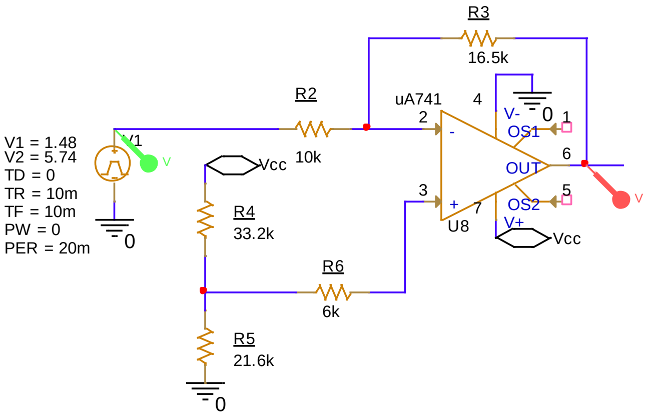

Figure 17 - Linear Voltage Domain Converter

The purpose of this circuit is to convert the input voltage that the comparator receives (between 1.48V to 5.74V) into a new domain of variation that is defined in the new range of 2V to 9V. To achieve this response, the following calculations were performed:

$$\frac{R3}{R2} = \frac{V_{O,max} - V_{O,min}}{V_{cd,max} - V_{cd,min}} = \frac{9 - 2}{5.74 - 1.643} = 1.643 \qquad [3]$$

For this result, the values of resistors that I chose are: R3 = 16.43kΩ and R2 = 10kΩ, as shown in the figure above. Going further into computing and explaining Vref, I chose to have a voltage divider structure formed by resistances R4 and R5, computed in equation [4],connected to the resistor R6, which has the role of assuring impedance matching between the two inputs of the amplifier. To compute R6, it is necessary to see which is the equivalent impedance in v_- input.

$$R6 = \frac{R3 \cdot R2}{R3 + R2} = \frac{16.43 \cdot 10}{26.43} = 6k\Omega \qquad [4]$$

Vref is computed as in the equation below:

$$V_{REF} = \frac{v_{O,min} + \frac{R3}{R2} v_{cd,max}}{1 + \frac{R3}{R2}} = \frac{2 + 1.643 \cdot 5.73}{1.643} = 4.324 \qquad [5]$$

The values of the resistances R4 and R5, that were used as a voltage divider to gain Vref without declaring a new voltage source, are calculated this way:

$$\frac{R4}{R4 + R5} 11 = 4.324 \rightarrow \frac{R4}{R4 + R5} = 0.393 \qquad [6]$$

The chosen value for R4 and R5 is 33.2kΩ and 21.5kΩ.

All the resistances computed above are providing the Linear Voltage Domain Converter in Figure 17 the necessary context to convert the input voltage into a domain of interval [2,9][V], as we can observe in the simulation below:

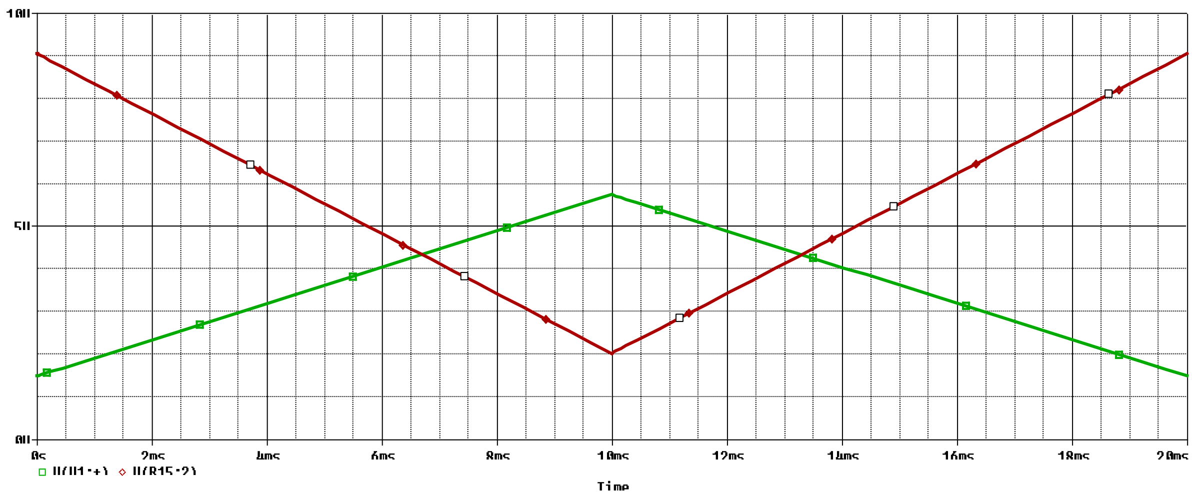

Figure 18 - Linear Voltage Domain Converter analysis

I will also note the fact that that for the representation in Figure 18, I used the same simulation profile as presented in Figure 10.

Trace Color	Trace Name	Y1	Y2
	X Values	0.000	10.000m
CURSOR 1,2	V(V1:+)	1.4800	5.7400
	V(U8:OUT)	9.0427	2.0166

Figure 19 - Values for analysis of Linear Voltage Domain Converter

2.2.4 Hysteresis Comparator

A hysteresis comparator is a positive feedback technique used in configurations to provide pre-determined comparator threshold. In my project, the use of this comparator is to convert the electric voltage that corresponds to 400ppm and 6000ppm into some voltage thresholds that will open or close the fan according to the requirement.

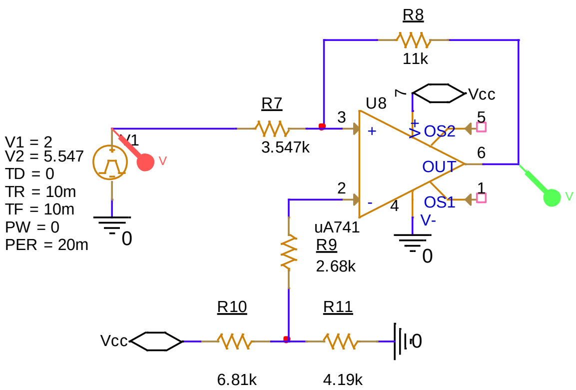

Figure 20 - Hysteresis comparator

As shown in Figure 20, the comparator receives at the input between 2V and 9V, these values being the low and high thresholds of the comparator, which will help to compute the resistances that compound the comparator. Until then, I need to clarify why we need 5.547V at the input of the comparator and how I made the conversion from the initial gas concentration range (400ppm to 9000ppm) to the required range (400ppm to 6000ppm), that ensures that the sensor's output voltage remains within the desired range to maintain system operation. This procedure implies a scaling down of the sensor's output to fit within the system's requirements. To achieve this goal, I converted the input info of the requirement into a graph that helped me compute the new range in which the sensor needs to operate in order to acquire the new working voltages.

The graph at Figure 21 shows the relation between the gas concentration vs. the sensor resistance: at 35kΩ we have a gas concentration of 400ppm, while at 5kΩ the concentration is 9000ppm. To compute the resistance that corresponds to 6000ppm, I firstly computed the slope of the graph:

$$m = \frac{5 - 35}{9000 - 400} = -0.00348837 \qquad [7]$$

$$R_{sensor\ new} = -0.00348837 \cdot (6000 - 400) + 35 = 15.46k\Omega \qquad [8]$$

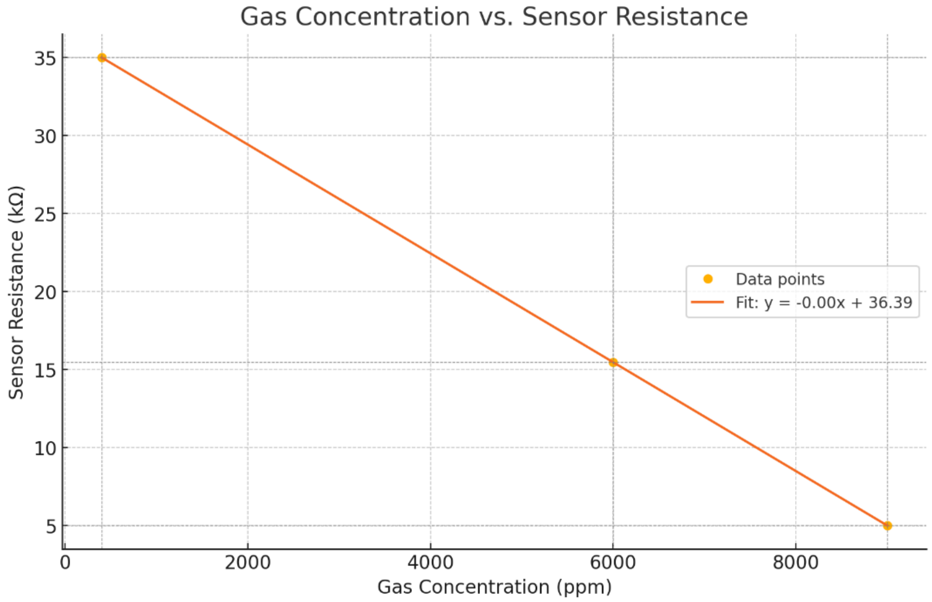

Figure 21 - Gas Concentration vs. Sensor Resistance

Having this information, we can conclude that the new sensor range varies between 35kΩ at 400ppm and 15.46kΩ at 6000ppm, which means that there is a need to compute the new threshold voltage for 6000ppm. Going back to the voltage divider at Figure 9, I will find the new voltage over the sensor to transfer it into the equation which computes the lower threshold of the hysteresis comparator.

$$v_{R_{sensor\ max\ new}} = \frac{R_{sensor\ max}}{R_{sensor\ max} + R1} \cdot Vcc = \frac{15.46}{15.46+32} \cdot 11 = 5.547V \quad [9]$$

$$v_{ThL} = -\frac{R7}{R8} V_{OH} + \left(1 + \frac{R7}{R8}\right) V_{REF} \quad [10]$$

$$v_{ThH} = -\frac{R7}{R8} V_{OL} + \left(1 + \frac{R7}{R8}\right) V_{REF} \quad [11]$$

$$\begin{cases} 2 = -\frac{R7}{R8} 11 + \left(1 + \frac{R7}{R8}\right) V_{REF} \\ 5.547 = -\frac{R7}{R8} 0 + \left(1 + \frac{R7}{R8}\right) V_{REF} \end{cases} \quad [12]$$

$$2 = -\frac{R7}{R8} 11 + 5.547 \quad [13]$$

$$\frac{R7}{R8} = \frac{3.547}{11} \quad [14]$$

From the last equation, I chose the values for the resistors: R7 = 3.547kΩ and R8 = 11kΩ. R9 is computed as the parallel connection between R7 and R8, giving the value R9 = 2.68kΩ.

Finally, I computed Vref, for which I designed a voltage divider, to whose resistances are:

$$Vref\left(1+\frac{3.547}{11}\right)=5.547 \quad [15]$$

$$\frac{R11}{R11+R10}11=4.19 \quad [16]$$

The values that I chose for the resistances are: R10 = 6.81kΩ and R11 = 4.19kΩ.

Now that this part of circuit is dimensioned, I will present the analysis (Figure 22) and its result (Figure 23). Note that this analysis has the same simulation profile as Figure 14 - Simulation profile of time domain analysis has.

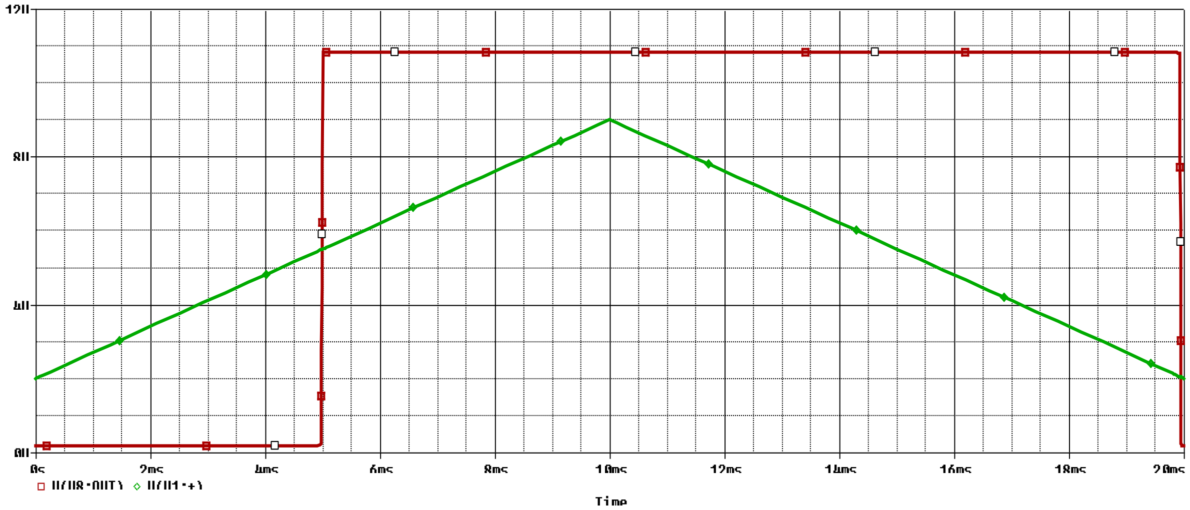

Figure 22 - Analysis of the hysteresis comparator

Trace Color	Trace Name	Y1	Y2
	X Values	0.000	4.9913m
CURSOR 1,2	V(U8:OUT)	183.815m	7.1589
	V(V1:+)	2.0028	5.4948

Figure 23 - Results of the analysis of the hysteresis comparator

By analyzing these results, we can observe that when the low threshold has the value of 2V the hysteresis starts and ands at the higher threshold of 5.49V, which means that the desired value is achieved and the hysteresis will help my system to work in the specified input parameters.

2.2.5 Led and Relay

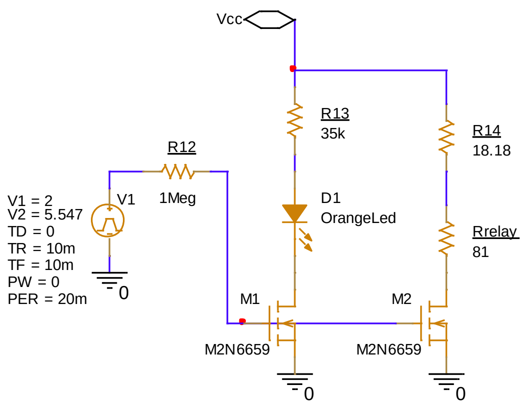

Figure 24 - Led and Relay Circuit

Relays typically require more current to operate than what a comparator can provide, so I needed transistor M3 to act as a current amplifier, allowing a low-current control signal to witch a higher-current relay. The second transistor came as a necessity, because the signal was slightly distorted for the relay when I only had one transistor, so I chose to add one of the same model, having the gate terminal in the same point as the first transistor, so that they operate simultaneously. In order to have functional transistors that are able to control the coil resistance of the relay, I placed a very large resistor (R12= 1MΩ), whose aim is to stabilize the gate voltage and to assure protection against voltage spikes.

Having an input voltage of 11V, I searched for the datasheet of a relay that can be supplied by this value. Due to the fact that I did not find a relay that works at this value of the Vcc, I chose a relay of 81Ω, which can be supplied at 12V, and I added a series resistance of value 18.18Ω, which I computed to be large enough to replace the lack of the 11V supplied relay.

In parallel with the relay branch I added the orange LED in series with a resistor of 35kΩ to make sure that the LED would not burn due to the high voltage. I computed resistor R13 as in the equation [17].

$$V_{CC} - V_{R13} - V_{DSsat} - V_{LED} = 0 \quad [17]$$

$$V_{DSsat} = V_{GS} - V_{Th} = 10 - 1.7 = 8.3V \quad [18]$$

$$V_{R13} = 11 - 8.3 - 2 = 0.7V \quad [19]$$

$$R13 = \frac{0.7}{20 \cdot 10^{-3}} = 35k\Omega \quad [20]$$

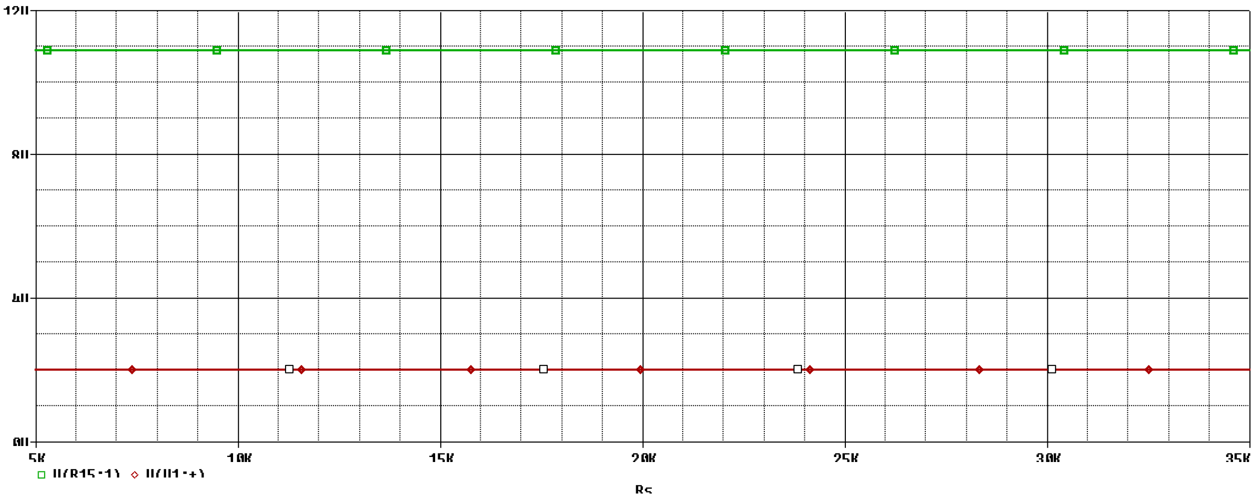

Figure 25 - Voltage on LED vs Voltage on Hysteresis Comparator

As it can be seen, the LED works as long as the Hysteresis comparator works => when the fan is on, the LED is on and vice versa, which means that the LED correctly signalizes if the concentration of carbon monoxide stays in the specified range or not. It can also be proven that the electric voltage over the LED is 2V, exactly as in the datasheet.

2.3 The functionality of the initial circuit

In the last subchapter, I presented the solution for all the parts of the circuit and showed that they work separately, but I will present in this subchapter that the circuit still works even with all the parts added together.

To acquire this objective, I will use the same DC analysis, which varies the resistance of the resistive sensor to prove the functionality of the circuit.

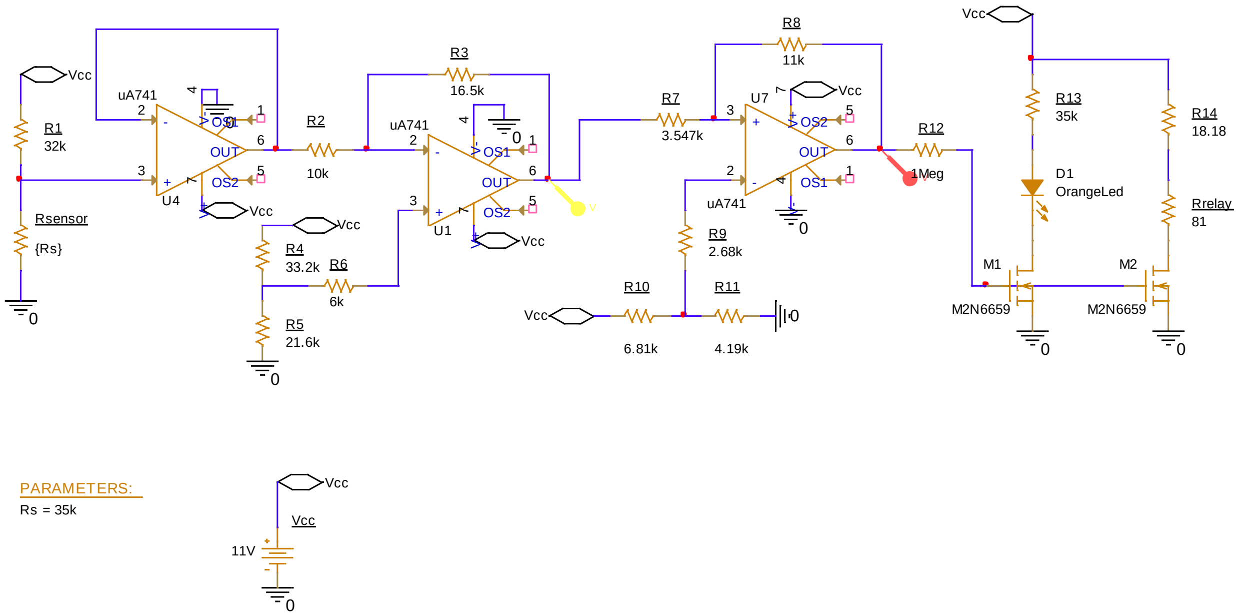

Figure 26 - Testing Domain Converter and Hysteresis

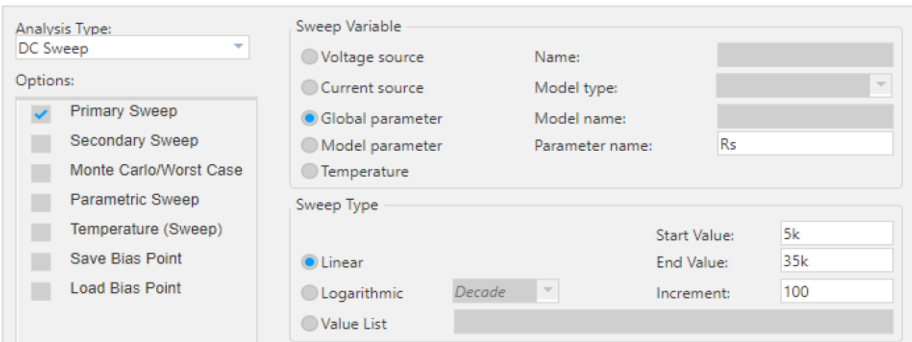

Figure 27 - Ascending resistance simulation profile

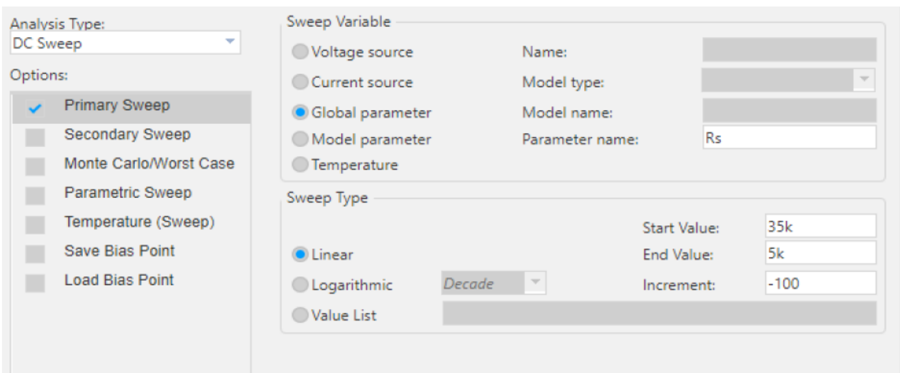

Figure 28 -Descending resistance simulation profile

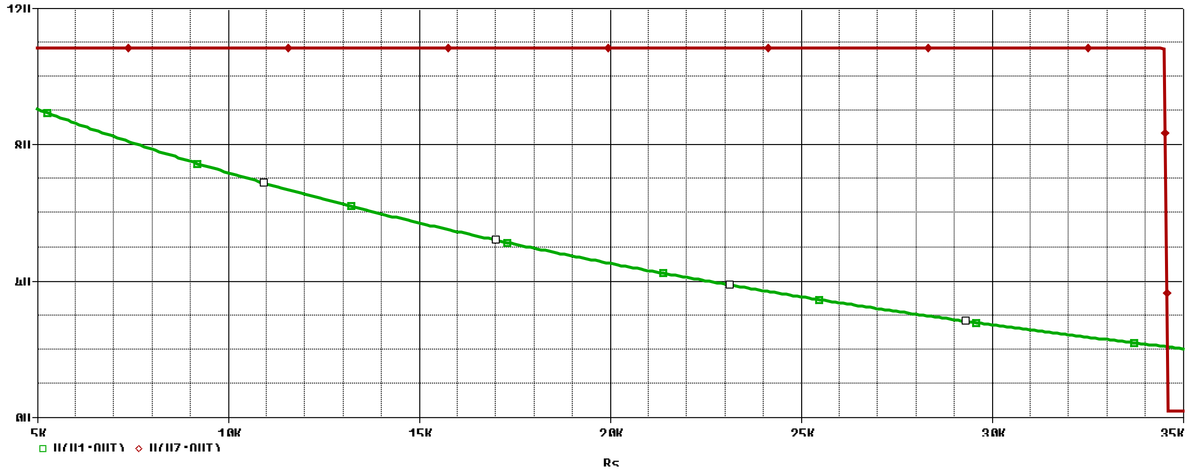

Figure 29 - Voltage domain and hysteresis simulation for high gas concentration

Trace Color	Trace Name	Y1	Y2
	X Values	34.581K	5.0262K
CURSOR 1,2	V(U1:OUT)	2.0626	9.0237
	V(U7:OUT)	2.1712	10.816

Figure 30 - Voltage vs Resistance values for high gas concentration

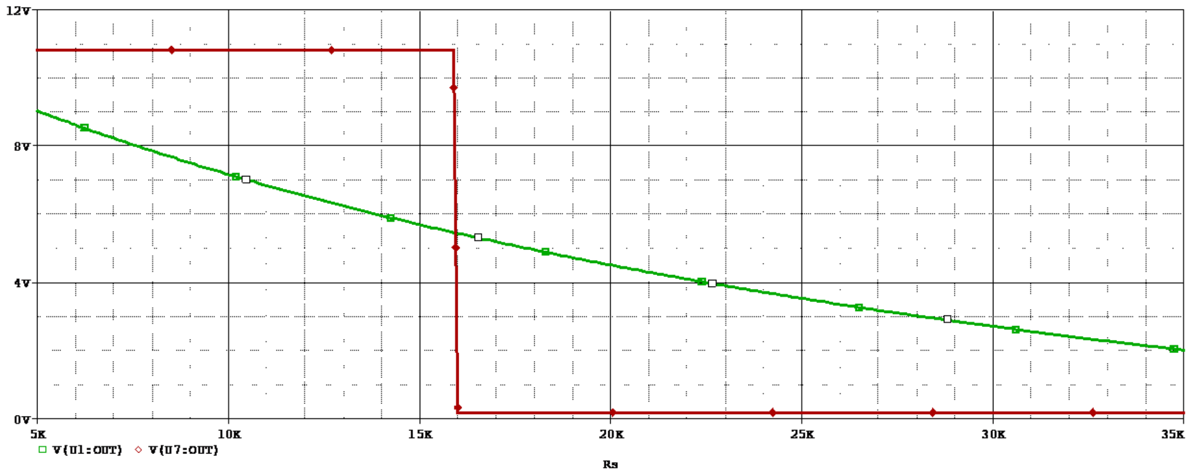

Figure 31 - Voltage domain and hysteresis simulation for low gas concentration

Trace Color	Trace Name	Y1	Y2
	X Values	35.000K	15.969K
CURSOR 1,2	V(U1:OUT)	2.0081	5.4464

Figure 32 - Voltage vs Resistance values for low gas concentration

In Figure 29 it can be shown that at low resistance (5kΩ), the gas concentration is high and the fan will be on until the gas concentration will be low and the resistance will have the highest value (35kΩ), while Figure 32 shows that between 35kΩ and 15.96kΩ the sensor detects the proper concentration of carbon monoxide in the enclosure, so that the fan is off, being turned on again when the resistance is again to loo low and the gas concentration is again unproper.

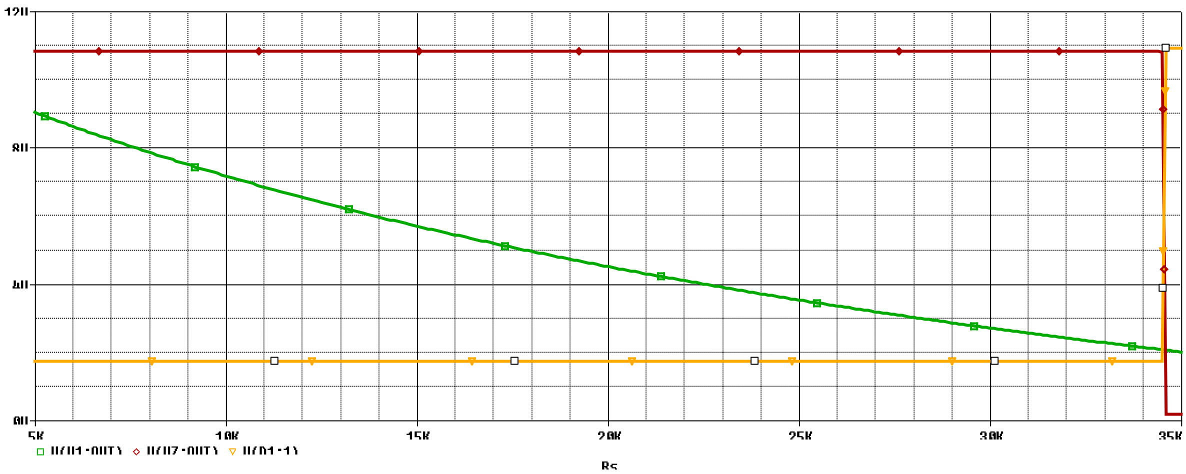

Figure 33 - LED vs hysteresis characteristic for high concentration of gas

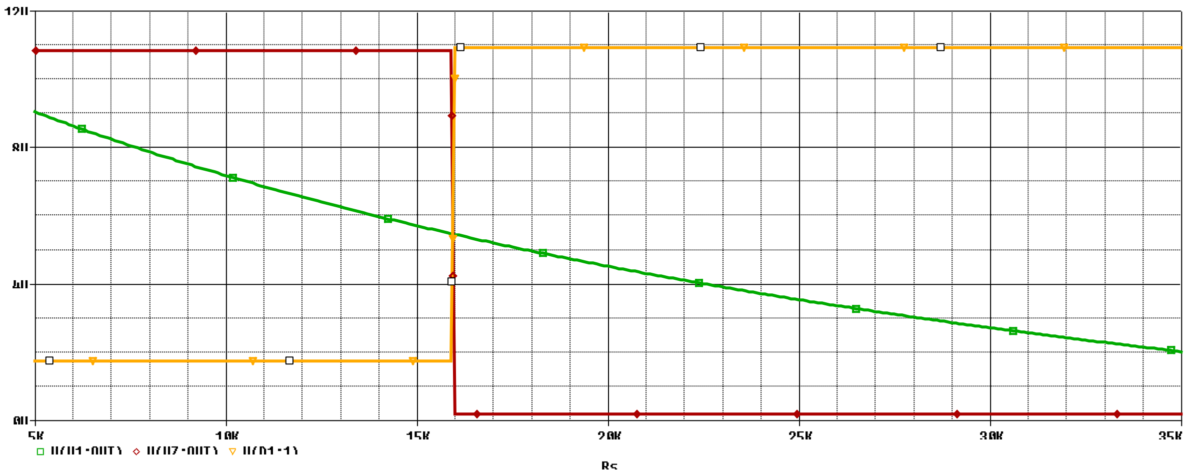

Figure 34 – LED vs hysteresis characteristic for low concentration of gas

It can be observed that the LED works as long as the sensor is on, being closed while the fan is working.

3 Standardized Circuit

3.1 General information

In order to give my circuit real-life functionality, I needed to standardize the values of the resistances, so that I will assure the client that my design is suitable for series production. To do so, I used Jansson, a very useful app in which I can choose the series of the resistances that I want to work it and introduce my calculations for achieving real life resistances. I worked with series E26, because it is a series with resistances of high accuracy and close tolerance requirements.

All of these being said, shows the standardized circuit.

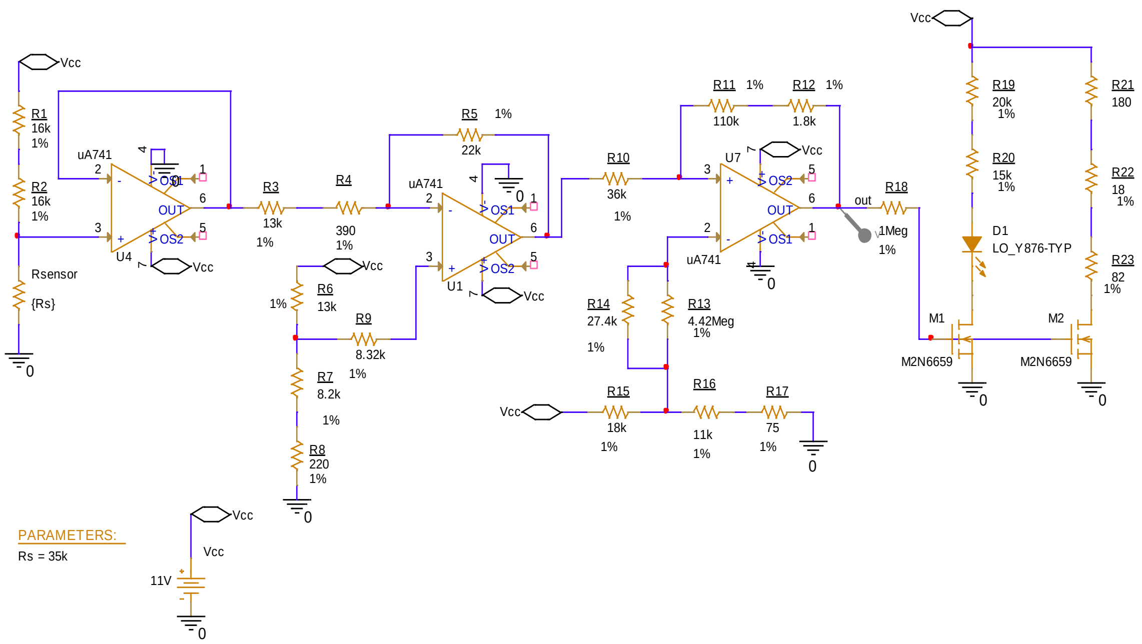

Figure 35 - Standardized circuit

A to demonstrate that the circuit keeps its functionality even after standardization, I will repeat the simulations in the Figure 29 and Figure 31, using Figure 33 and Figure 34 as simulations profiles, just like above.

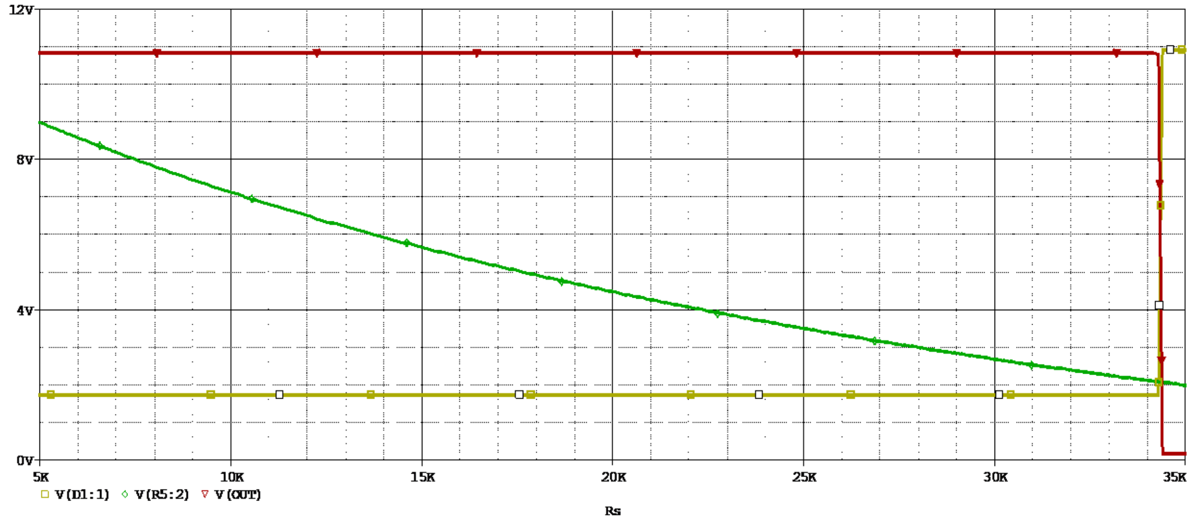

Figure 36 - Voltage domain, hysteresis and LED simulation for standardized circuit, high concentration of gas

Trace Color	Trace Name	Y1	Y2
	X Values	35.000K	5.0000K
CURSOR 1,2	V(D1:1)	10.907	1.7377
	V(R5:2)	2.0014	8.9854

Figure 37 - Interpretation of the voltage domain and hysteresis for standardized circuit, high concentration of gas

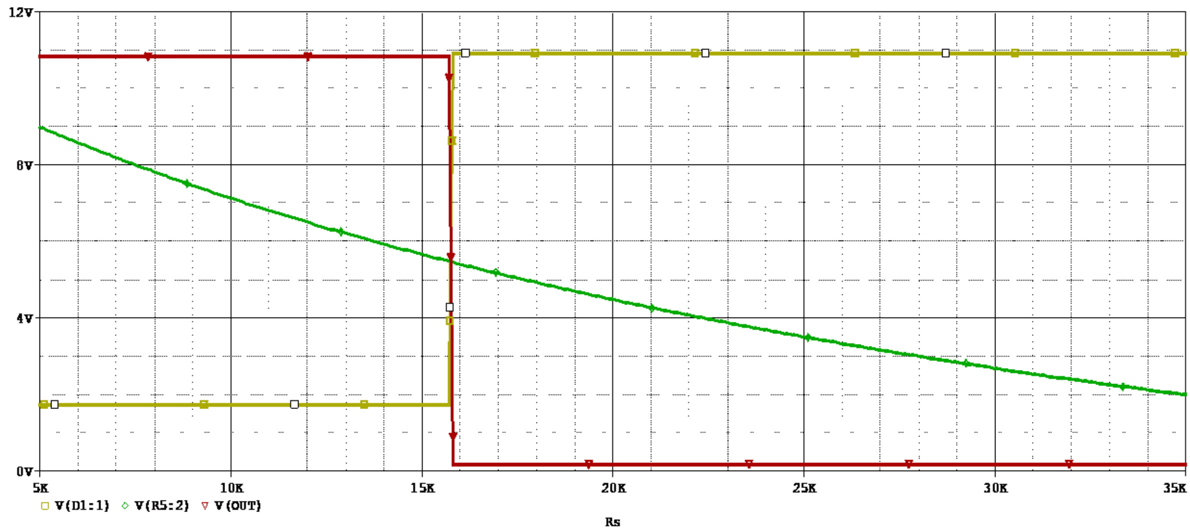

Figure 38 - LED's characteristics in comparison with the hysteresis for standardized circuit, low concentration of gas

Trace Color	Trace Name	Y1	Y2
	X Values	15.741K	34.900K
CURSOR 1,2	V(D1:1)	5.5159	10.907
	V(R5:2)	5.4651	2.0014

Figure 39 - Interpretation of the LED's characteristics in comparison with the hysteresis for standardized circuit, low concentration of gas

Comparing the results of the standardized circuit and the initial one, we can observe that the non-standardized circuit is has an efficiency with 0.58% higher than the standardized one.

3.2 Worst Case analysis

In order to prepare this analysis, I tried to change different types of tolerances so that I obtained an output file with as many unchanged resistances during worst case. Firstly, I started with a tolerance of value 5%, then 1% and the 0.1%, and I found out that 1% is the best tolerance for my resistances.

Device	MODEL	PARAMETER	NEW VALUE	
R_R13	R_R13	R	1	(Unchanged)
R_R5	R_R5	R	.99	(Decreased)
R_R23	R_R23	R	1	(Unchanged)
R_R16	R_R16	R	1	(Unchanged)
R_R9	R_R9	R	1	(Unchanged)
R_R20	R_R20	R	1	(Unchanged)
R_R8	R_R8	R	1	(Unchanged)
R_R1	R_R1	R	1	(Unchanged)
R_R18	R_R18	R	1	(Unchanged)
R_R4	R_R4	R	1	(Unchanged)
R_R22	R_R22	R	1	(Unchanged)
R_R2	R_R2	R	1	(Unchanged)
R_R19	R_R19	R	1	(Unchanged)
R_R3	R_R3	R	1	(Unchanged)
R_R6	R_R6	R	.99	(Decreased)
R_R15	R_R15	R	1	(Unchanged)
R_R7	R_R7	R	1.01	(Increased)
R_R17	R_R17	R	1	(Unchanged)
R_R10	R_R10	R	1	(Unchanged)
R_R11	R_R11	R	.99	(Decreased)
R_R12	R_R12	R	1	(Unchanged)
R_R14	R_R14	R	1	(Unchanged)

Figure 40 - Resistors' s tolerances while performing worse case analysis

In Figure 40 I will present the worse case analysis, on which graph we can see that the circuit is stable due to the fact that it does not varies in time. The simulation profile for this analysis is presented in Figure 41.

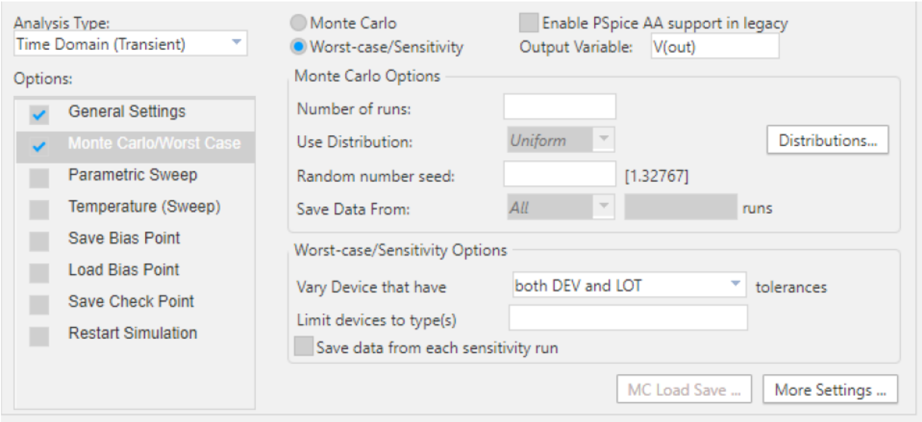

Figure 41 - Simulation profile for worse case analysis

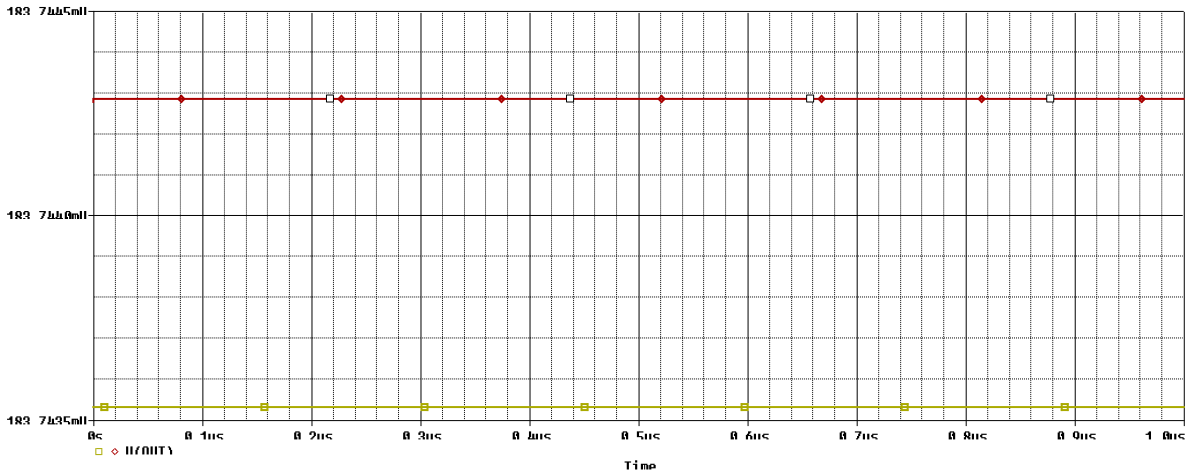

Figure 42 - Worse case analysis

3.3 Monte Carlo analysis

The Monte Carlo simulation is used to model the probability of different outcomes in a process that cannot easily be predicted due to the intervention of random variables. It is a technique used to understand the impact of risk and uncertainty.

In PSpice, this analysis helps in understanding how variations in resistors, capacitors, inductors and other component tolerances affect the overall circuit performance. This is crucial for ensuing reliability and robustness in mass production. All these being said, I will show the simulation profile in Figure 43 and the results of the analysis in below.

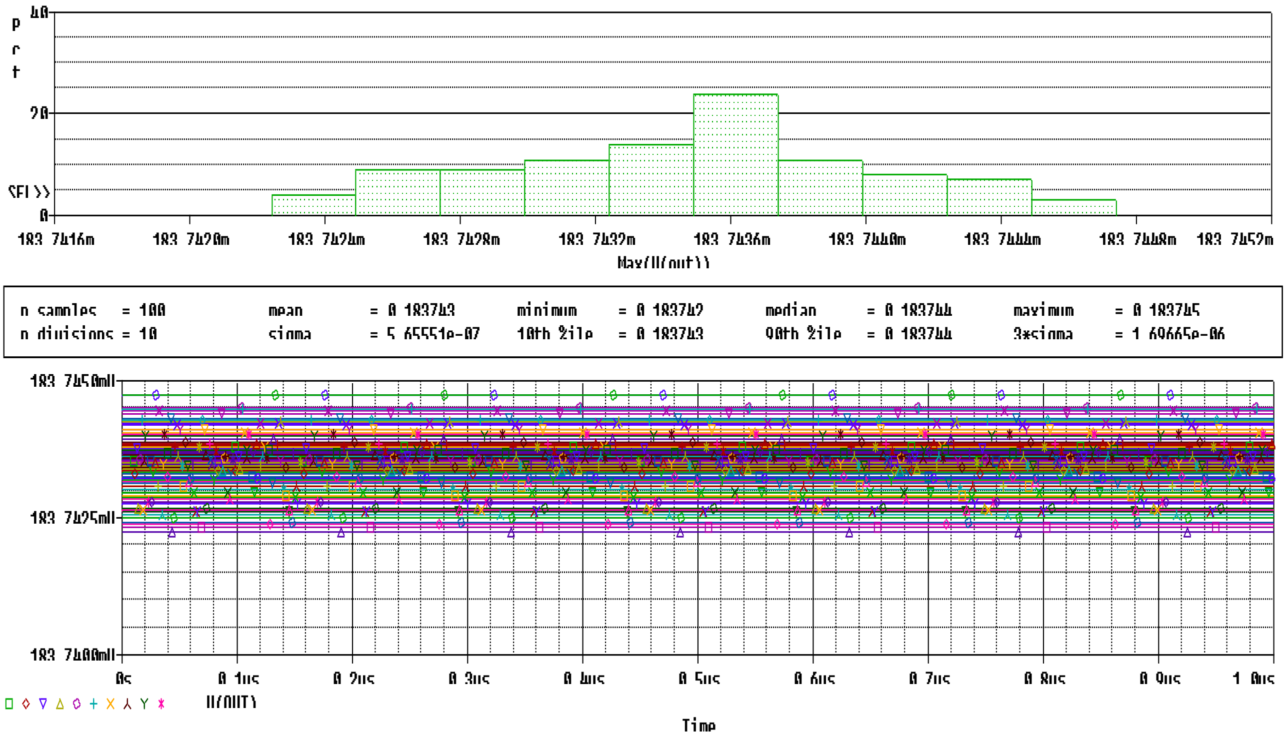

Figure 43 - Monte Carlo and Performance analysis

The histogram at the top corresponds to the Performance analysis, which shows the distribution of a performance metric of the circuit. During this analysis, the modeling and simulation of circuits with amplifiers are vary many parameters to achieve an improvement in the stabilization. The function that I used in order to perform this analysis was V(out), so that the histogram will show how my output voltage varies at 100 samples, particularly changing the tolerances of resistances(which have a tolerance of 1%), to observe the impact of the output voltage. The output voltage follows a roughly normal distribution, indicating that most of the output voltage values cluster around a central value, with fewer occurrences as it moves away form the center. This distribution helps us visualize the variability and stability of the output voltage due to the 1% tolerance in resistances, which shows that the chose of this value for the tolerance was the right choice.

Having a look over the Monte Carlo analysis, the one below the histogram, we can visualize the same result as the one specified above, but this time in a different visualization, with different markers and colors that represent different test conditions for components, the same, as the tolerance is changed, varying the output voltage. This variation does not exceed 0.0025mV, resulting in understanding that the circuit is stable for the chosen value of the tolerances.

4 Bill of materials

Component	Quantity	Value	Tolerance	Model	Price (RON)
LedOrange	1	/	1%	LO_Y876-TYP	1,1
M1,M2	2	/	1%	M2N6659	
Rsensor	1	/	1%	/	
R1,R2	2	16 [kΩ]	1%	560112110032	0,92
R3,R6	2	13 [kΩ]	1%	ESR03EZPF1302	1,39
R4	1	390 [Ω]	1%	CRG0402F390R	0,46
R5	1	22 [kΩ]	1%	MCS04020C2202FE000	0,46
R7	1	8.2 [kΩ]	1%	CRCW06038K20FKEC	0,46
R8	1	220	1%	MCS04020C2200FE000	0,46
R9	1	8.32[kΩ]	1%	RN73H2BTTD8351F100	1,94
R10	1	36[kΩ]	1%	ESR03EZPF3602	0,693
R11	1	110[kΩ]	1%	MCT06030C1103FP500	0,46
R12	1	1.8[kΩ]	1%	CRCW12101K80FKEA	1,01
R13	1	4.42[MΩ]	1%	CRCW04024M42FKED	0,46
R14	1	27.4[kΩ]	1%	CRCW040227K4FKEDC	0,46
R15	1	18[kΩ]	1%	560112116050	0,505
R16	1	11[kΩ]	1%	560112110123	0,46
R17	1	75[Ω]	1%	CRCW080575R0FKEAHP	0,693
R18	1	1[MΩ]	1%	CRCW12061M00FKEC	0,46
R19	1	20[kΩ]	1%	560112110029	0,46
R20	1	15[kΩ]	1%	560112110127	0,46
R21	1	180[Ω]	1%	SFR03EZPF1800	0,505
R22	1	18[Ω]	1%	CRGH0805F18R	0,508
R23	1	82[Ω]	1%	ERJ-P06F82R0V	0,599
U1,U4,U7	3	/	/	UA741CP	7,44
Total	**21,26 RON**				

5 Conclusion

In conclusion, I designed a functional system that uses a resistive gas sensor to maintain the concentration of carbon monoxide within a specified range in an enclosure. The use of reliable electronic components is evident in *Bill of materials*, where I determined that the implementation of my design would cost only 21.26 RON, making it extremely affordable for mass production. Although I opted for a less performant but cost-effective operational, this choice contributes to the overall practicality and cheapness of the design.

One drawback of my design is that the LED remains on as long as the gas concentration in the enclosure is appropriate. This means that, in everyday use, the LED will be on most of the time, potentially leading to quicker degradation. Despite this minor inconvenience, I believe that my design is reliable and efficient for controlling the concentration of carbon monoxide in an enclosure.

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